

The new, all-electric Beemer

The latest buzz is the BMW i3's launch in July 2013. The model is expected to be priced much higher than its counterparts from other brands. So what makes the all-electric i3 so special? One is the Life Module—a body constructed of special material. Then, there is the completely battery-powered electric engine and other sustainability features. Apart from all this, there is also the innate intelligence of the vehicle, enhanced with seamless smartphone integration and remote control via the BMW ConnectedDrive application.

Inside: The electric engine generates an output of 125 kW/170 hp (and a maximum torque of 250 Newton-metres) by transferring a spontaneous onset of power via a single-stage gearbox to the rear wheels. The e-motor of the car is powered by a special battery housed below the chassis. This special, high-voltage lithium-ion battery typically requires recharging only once in two to three days, i.e., after 130 to 160 kilometres. It also includes an intelligent heating-cooling system, which ensures that its energy output is not deeply affected by temperature fluctuations. Intelligent electronics, complete use of LED lights, smart air-conditioning and eco driving modes help conserve energy further.

The car features a passenger cell made of carbon fibre-reinforced plastic (CFRP) and an aluminium chassis carrying the battery, motor and suspension. This new material not only compensates for the weight of the battery but also gives a lower centre of gravity for improved driving dynamics and safety. The i3 features advanced in-car electronics including connectivity to the outside world and two displays that are capable of high-definition 3D video.

The car connects to your smartphone using the ConnectedDrive navigation system, which has several useful features. One such feature is the dynamic range indicator that takes into account factors such as battery charge, driving style, topographic conditions and current traffic situation, to inform the driver about how long the battery will last, the nearest charging station (which may even be reserved in advance), how long the battery needs to be charged to complete the journey, etc.



Mind-controlled mobile devices

Samsung's Emerging Technology Lab along with Roozbeh Jafari's team at the University of Texas in Dallas is developing a device that would allow users to control smartphones and tablets with their brain. The device, like many forerunning experiments of its kind, uses electroencephalography (EEG) as the basis for 'reading' the mind. While Samsung has no specific plans to develop a mind-controlled device yet, it has seen hopeful results in the experiments. According to recent media reports, users were able to perform an action with their mind at a speed of one action per five seconds, with an accuracy rate of 80 per cent to 95 per cent. If this technology becomes reality, it could be very useful for people with motor-control problems and other impairments.

Inside: The research is based on the team's observation of well-known brain activity patterns that occur when people are shown repetitive visual patterns. The device uses an EEG cap that the users have to wear to control a Samsung Galaxy Note 10.1. When they concentrate on an icon blinking at a distinctive frequency, the cap detects the brain activity pattern and converts it into an instruction to launch an application, select a specific person out of a list of contacts or a song from a certain playlist, or even turn on or turn off a mobile device. Although full of wires, the EEG cap does not use a liquid bridge for the sake of comfort. As a result of this, the resulting signals are a bit weak. So the team is trying to come up with better signal processing methods to extract the correct information required to control a device from relatively weak EEG signals and convert these into inputs for the device. They are also hoping to develop easy-to-use EEG sensors and to make the whole system more comfortable.

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